

9.1.9.5

Reprezentări

Coduri de reprezentare a nr. cu semn
 întregi
 $x_{(10)} \rightarrow x_{(2)}$

subunitate
 $0, x_{(10)} \rightarrow 0, x_{(2)}$

n biți disponibili

$$[x]_D = \begin{cases} \begin{array}{|c|c|} \hline 0 & |x| \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & |x| \\ \hline \end{array} \end{cases}$$

$$[0, x]_D = \begin{cases} \begin{array}{|c|c|} \hline 0 & |x| \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & |x| \\ \hline \end{array} \end{cases}$$

$$[x]_i = \begin{cases} \begin{array}{|c|c|} \hline 0 & |x| \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & [x]_i \\ \hline \end{array} \end{cases}$$

$$[0, x]_i = \begin{cases} \begin{array}{|c|c|} \hline 0 & |x| \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & |x| \\ \hline \end{array} \end{cases}$$

$$[x]_C = \begin{cases} \begin{array}{|c|c|} \hline 0 & |x| \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & [x]_C \\ \hline \end{array} \end{cases}$$

$$[0, x]_C = \begin{cases} \begin{array}{|c|c|} \hline 0 & \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & \\ \hline \end{array} \end{cases}$$

$$x = 245_{(10)} = \overline{F5}_{(16)} = 1111 \ 0101_{(2)} \quad n = 8 \text{ biți}$$

$$[x]_D = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ \hline \end{array}$$

$$[x]_i = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ \hline \end{array}$$

$$[x]_C = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ \hline \end{array}$$

$$x = -125_{(10)}$$

$$n = 8 \text{ biți}$$

$$\text{bază} = 8$$

$$125_{(10)} = 175_{(8)} = 001 \ 111 \ 101_{(2)}$$

$$110 \ 000 \ 010$$

$$125 : 8 = 15 = 1 = 0$$

571

$$[x]_D = \underline{1111111011}$$

$$[x]_i = \underline{110101010110}$$

$$[x]_C = \underline{1101010111}$$

$$z = 0,125_{10} = 0,1_{(8)} = 0,001_{(2)}$$

$$\begin{array}{r} 125 \cdot \\ 8 \\ \hline 1000 \\ 0 \cdot \\ 8 \\ \hline 0 \end{array}$$

$$[z]_D = 0,00100000$$

$$[z]_i = 0,00100000$$

$$[z]_C = 0,00100000$$

Operații în cod complementar

$$\begin{array}{r} [S_1] \quad \dots \\ [S_2] \quad \dots \\ \hline [S_n] \quad \dots \end{array}$$

Dacă $(S_1 = S_2) \neq S_n$ atunci avem depășire

$$[x]_C \oplus [y]_C = \begin{array}{r} 01110101 \oplus \\ 10000011 \\ \hline 11111000 \end{array} \quad \text{nu avem depășire}$$

$$[x]_C - [y]_C = [x]_C \oplus [-y]_C = \begin{array}{r} 01110101 \oplus \\ 01111101 \\ \hline 11110010 \end{array} \quad \text{avem depășire}$$

$$[t]_C - [z]_C = [t]_C \oplus [-z]_C = \begin{array}{r} 11100000 \oplus \\ 11110000 \\ \hline 111010000 \end{array} \quad \begin{array}{l} \text{nu avem} \\ \text{depășire} \\ (1=1=1) \end{array}$$

Reprezentarea în virgulă fixă

$m = 8$ bit:

$$x = 14_{(10)} = 32_{(16)} = 1110,0101_{(2)}$$

$$= \boxed{0 \mid 1 \mid 1 \mid 0 \mid 0 \mid 1 \mid 0 \mid 1}$$

$$14 = 3 = 0$$

23

$$0,32 \frac{1}{4}$$

$$\begin{array}{r} 1,28 \\ \hline 4 \end{array}$$

$$\begin{array}{r} 1,12. \\ 4 \end{array}$$

$$\underline{0,48}$$

$$y = -54,3_{(10)} = -66,23_{(8)} = \underset{\text{xxx}}{-110110}, \underset{\text{xx}}{010011}_{(2)} = \boxed{1111010101010101}$$

Reprezentarea în virgulă mobilă, simplă precizie (32 b)

- mantisă subunitară normalizată

E-exponent

$$C = E + Q$$

$$Q = 2^7 - 1 = 127$$

 $0, 1, \dots$

- mantisă între 1 și 2

$$14, 32_{(10)} = 32, 11_{(4)} = 1110, 0101_{(2)} = 0, 11100101_{(2)} * 2^4$$

$$E = +4$$

$$C = 4 + 2^7 - 1 =$$

$$= 2^7 + 3 = 1000 \ 0000 + 11 =$$

$$= 1000\ 0011(2)$$

S $C(8)$ $M(23)$

$$14, 32_{(10)} = 32, 110132232$$

$$0,32 \cdot$$

$$\overline{1,28}_4$$

$$\begin{array}{r} 1,12 \\ \underline{4} \end{array}$$

$$\underline{0.48}$$

$$0,48 \cdot 4$$

$$\frac{1,92}{4}$$

3,68

2.7.

$$\begin{array}{r} 0,72 \\ \hline \end{array}$$

2,88
4

3, 5 2

2, 0